IMSE 785

Big Data Analytics

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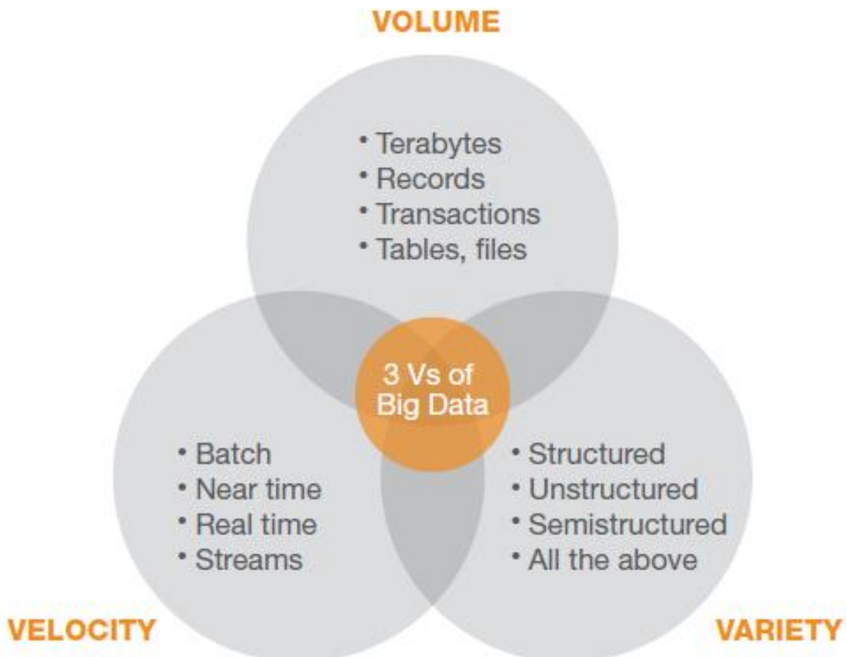
Lecture 2. Big Data Strategy

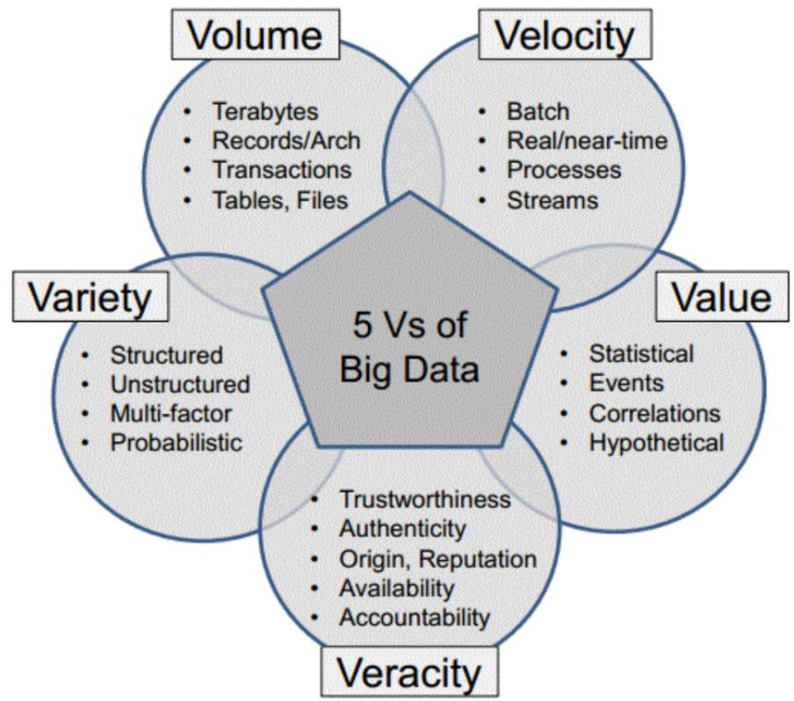
- When to Consider a Big Data Solution
- Who are end users of big data analytics?
- Who do you need in the <u>analytics team</u>
- What <u>Big Data Platform</u> should be used?
- What <u>Application Development Platform</u> should be used?
- <u>Enterprise Integration</u>: in house or cloud?

(source: chap 2 2012 understand Big Data from IBM)

Defining Big Data via the Three Vs



Source: Table 1. 2011 TDWI report sponsored by SAS



http://iihtofficialblog.blogspot.com/2014/07/5-vs-of-hadoop-big-data.html

When to Consider a Big Data Solution

- Big data solutions should be considered when:
  - Big Data solutions are good for raw structured, semistructured, and unstructured data
  - All data rather than a sample of it need to be analyzed
  - Iterative and exploratory analysis done on business measure are not predetermined

Questions to ask before switching to Big Data Analytics

- Is the switching to big data analytics appropriate for the task at hand?
- Can you see a Big Data platform complementing what you currently have in place for analysis?
- Is Big Data well suited for solving information challenges that don’t natively fit within a traditional relational database approach?

Case Study 1: Call Center Mantra

- Call centers would like to detect emerging problems to assistant CSR (customer service representatives) to address what's going on with lower latency
- An in-motion analytics (streams) means that you need to convert voice into text or employ voice analysis
- See https://www.youtube.com/watch?v=XWRsxy8SbzY&t=2s
- How do you capture and detect loyalty decay?

Case Study 2. Fraud Detection Pattern

- Fraud detections take place in many transaction-based or claims-based environments, such as online auctions, insurance claims, underwriting entities, etc.
- Challenges of traditional IT: what needs to be stored, lack of IT resources to run analyses, speed of modeling, analysis, and detection
  - Only 20% of data is analyzed

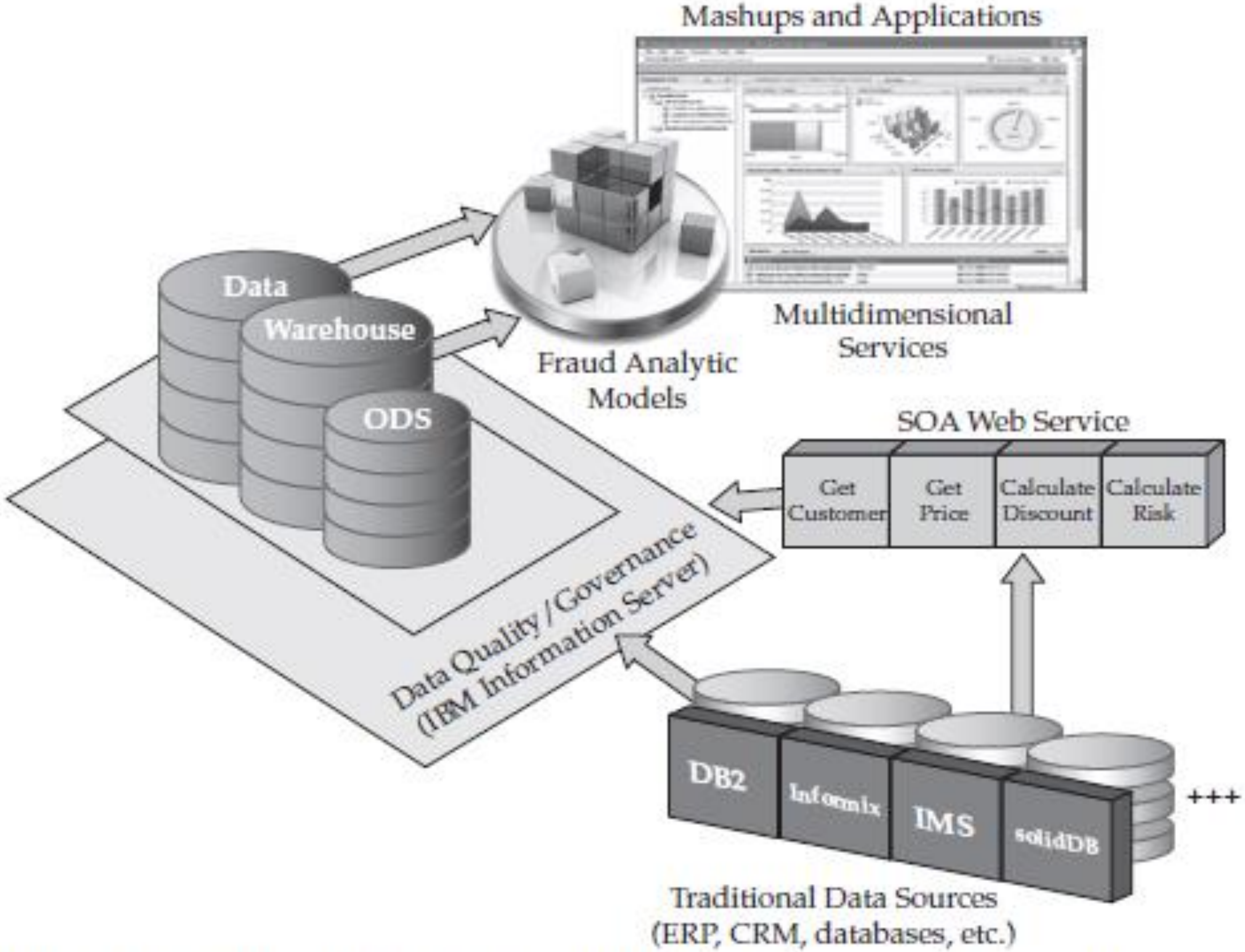


Figure 2-1 *Traditional fraud detection patterns use approximately 20 percent of available data.*

Source: 2012 IBM Understanding Big Data

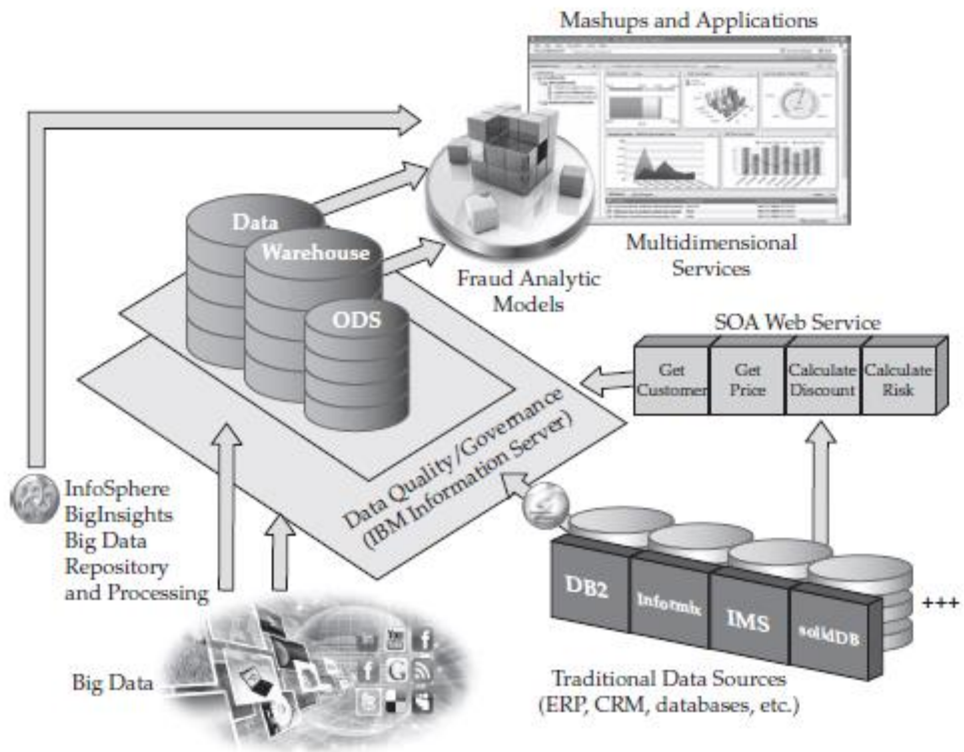


Figure 2-2 *A modern-day fraud detection ecosystem synergizes a Big Data platform with traditional processes.*

With streams, the fraud detection model can be applied when transaction are happing.

Source: 2012 IBM Understanding Big Data

Case 3. Detecting Pending Malfunction Machines

- Budweiser has installed sensors to detect motor sounds not detectable by humans
- Signal from soundwaves help detect pending motor failures before they take place
- Machine learning algorithms run in the cloud where data rests
- The goal is to reduce downtime to zero

Analytic team and Users

- Analytic team: IT specialists, Data Scientists
- Users: people in functional departments e.g. CRS, claim specialists.
- Should the users in each functional department be a part of the analytic team?

Components Under Hadoop

- **Pig™ (Yahoo)**: A high-level data-flow language and execution framework for parallel computation.
- **Hive™(Facebook)**: A data warehouse infrastructure that provides data summarization and ad hoc querying.
- **HBase™**: A scalable, distributed database that supports structured data storage for large tables.
- **Mahout™**: A Scalable machine learning and data mining library.

- Others

(source: http://hadoop.apache.org/ read: 2012 Understanding big data chap 4)

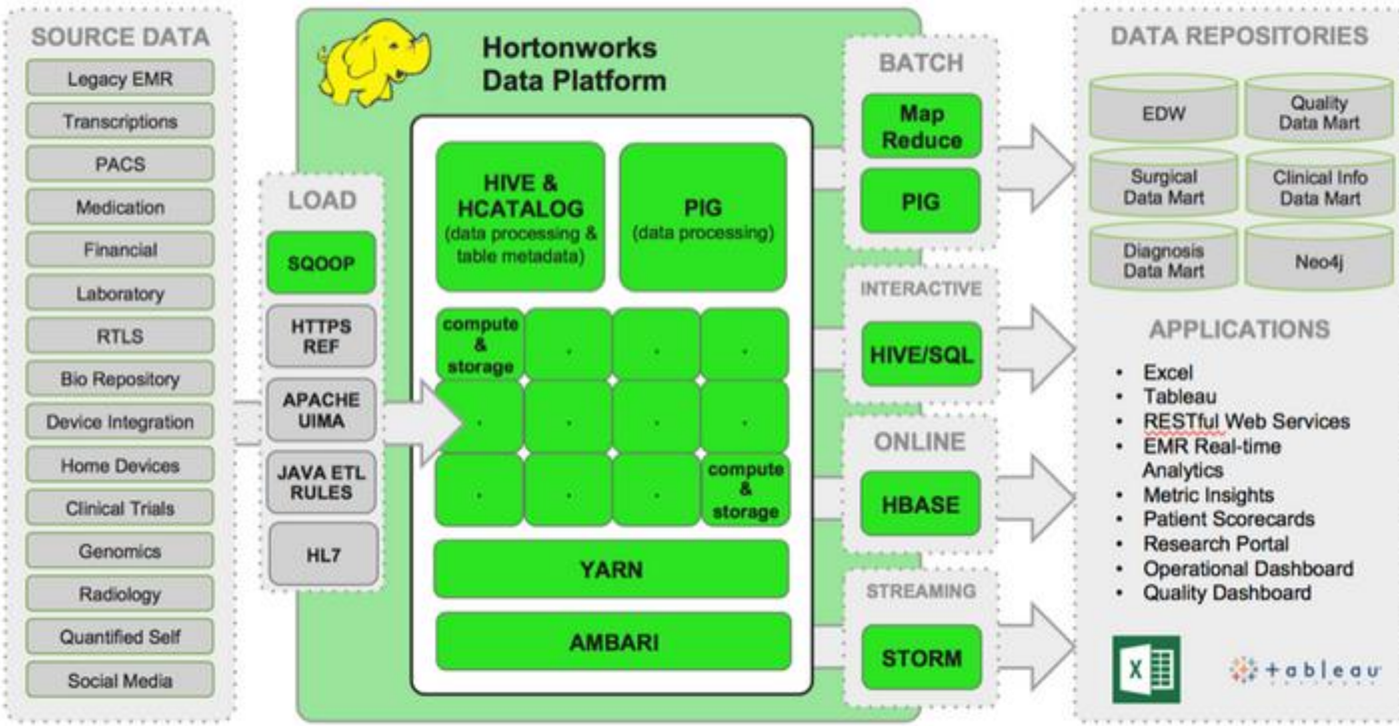
Cloud Computing

- Where do you implement your IT needs?

- Answer: three ways of cloud computing
  - SaaS (Software as a Service)
  - PaaS (Platform as a Service)
  - IaaS (Infrastructure as a Service)
- Video link (7 minutes)

http://www.youtube.com/watch?v=SgujaIzkwrE

Case Study 4: Big Data in Healthcare



Source: http://hortonworks.com/blog/modern-healthcare-architectures-built-with-hadoop/

Your turns

- Please identify various parts (source, temp storage, long-term storage, analysis/applications) of Enterprise Computing System in the health care big data architect (Note Hortonworks is the name of a company which help install Hadoop)
- How do you use the same architect for the beer production example?

Data Source

Sources of Healthcare Data

Source data comes from:

- Legacy Electronic Medical Records (EMRs)
- Transcriptions
- PACS
- Medication Administration
- Financial
- Laboratory (e.g. SunQuest, Cerner)
- RTLS (for locating medical equipment & patient throughput)
- Bio Repository
- Device Integration (e.g. iSirona)
- Home Devices (e.g. scales and heart monitors)
- Clinical Trials
- Genomics (e.g. 23andMe, Cancer Genomics Hub)
- Radiology (e.g. RadNet)
- Quantified Self Sensors (e.g. Fitbit, SmartSleep)
- Social Media Streams (e.g. FourSquare, Twitter)

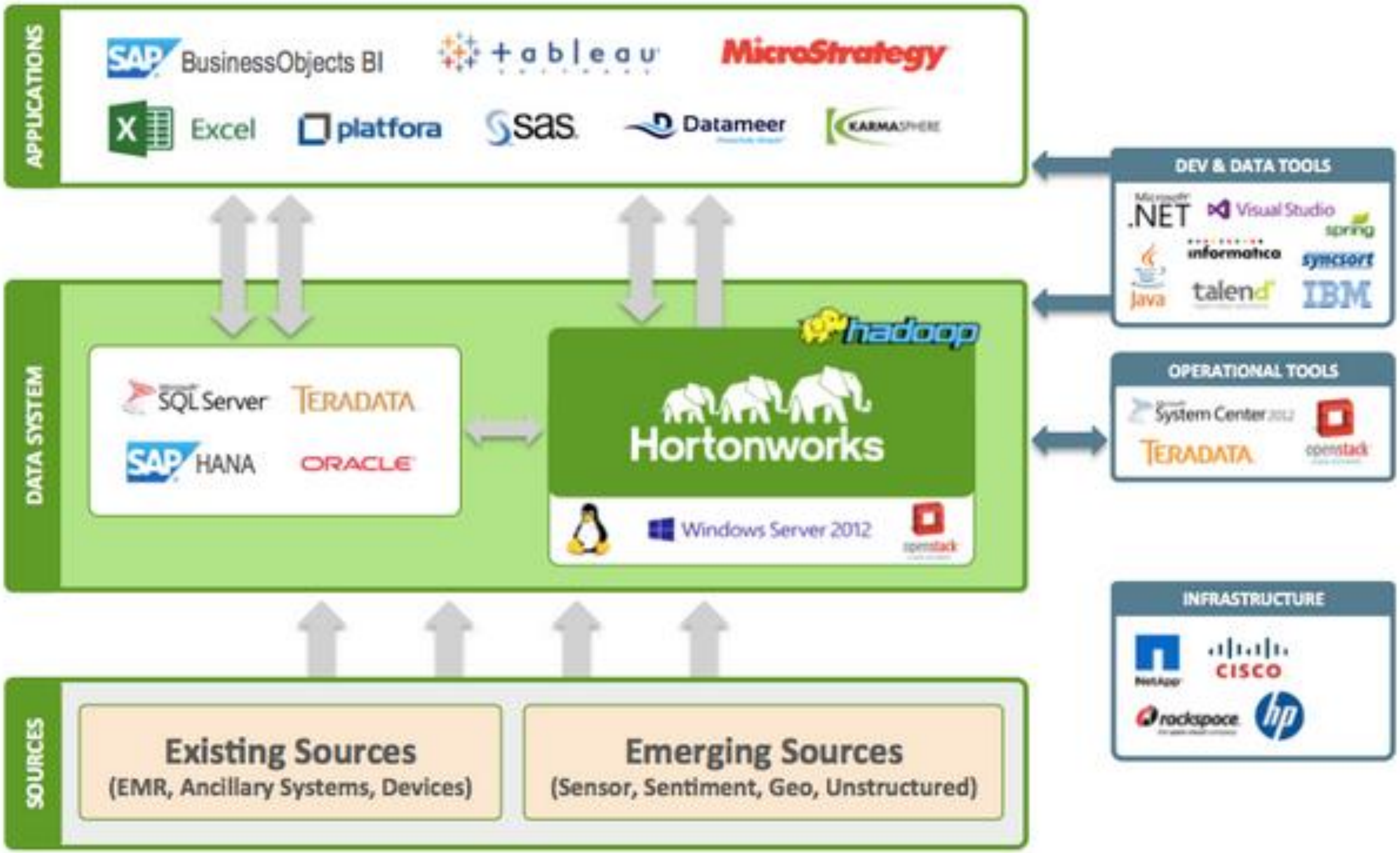
Hadoop Outputs

- Enterprise data warehouse
- Quality data mart
- Surgical data mart
- Clinical info data mart
- Diagnosis data mart
- Neo4j graph database

Analysis Tools

- Microsoft Excel
- Tableau
- RESTful Web Services
- EMR Real-time analytics
- Metric Insights
- Patient Scorecards
- Research Portals
- Operational Dashboard
- Quality Dashboards

Possible Data/Analysis Integration



Assignment 2. (10 points)

- Consider the application you chose in Assignment 1
- Search for the vendors (e.g. Hortonwork) for Hardoop
- Identify two vendors that you would like to compare
- Choose a vendor that may fit best for your company
- Your work will be graded on how thorough and appropriate the criteria and comparison are.